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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N°_7 – Réseaux I

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Niveau : L3

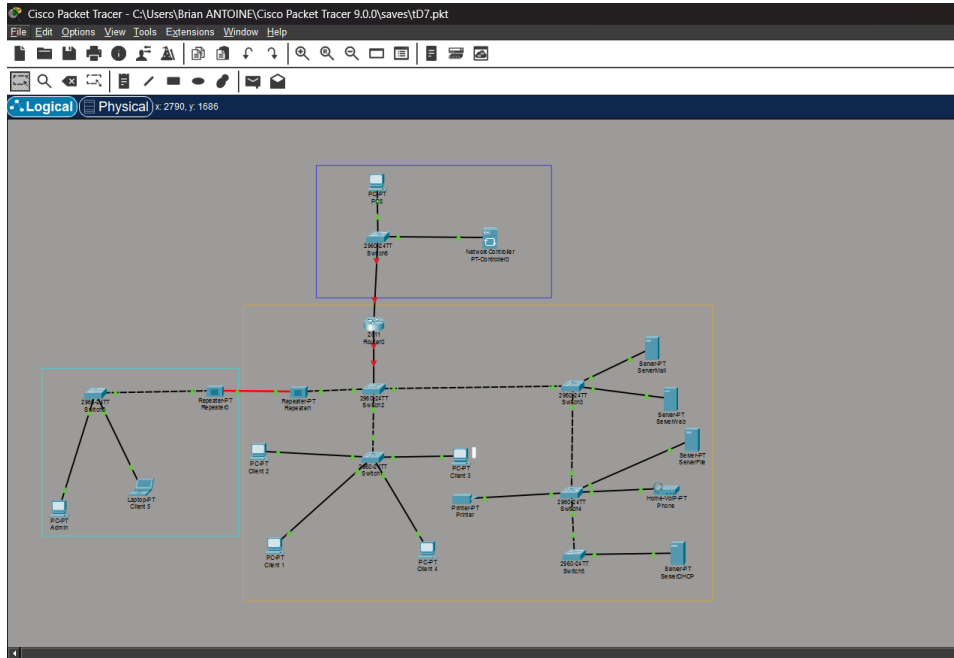
Date: 13/12/2025

L'objectif de ce TD est de :

1. Activer Telnet en mode sécurisé
2. Créer plusieurs niveaux d'utilisateurs
3. Restreindre l'accès Telnet via ACL
4. Journaliser les connexions
5. Mettre en place une bannière légale
6. Tester Telnet depuis plusieurs VLAN
7. Superviser les sessions actives
8. Configurer timeouts et protections
9. Configurer SSH avec clés RSA 2048/4096 bits
10. Mettre en place différents niveaux d'utilisateurs (privilèges 1, 5, 15)
11. Restreindre l'accès SSH via ACL
12. Activer SSH version 2 (obligatoire)
13. Configurer des timers de sécurité
14. Activer protection contre brute-force
15. Journaliser tentatives réussies/échouées
16. Superviser sessions SSH actives
17. Durcir l'équipement

Travaux dirigés

- **Reproduction de la topologie**



A. Configuration Telnet (partie non sécurisée)

➤ Configuration du routeur

The screenshot displays a Cisco IOS command-line interface. At the top, there are tabs for 'Physical', 'Config', 'CLI', and 'Attributes', with 'CLI' currently selected. The main window shows the following text:

```
States and local country laws governing import, export, transfer and
of Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wll/export/crypto/toccl/statg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249556K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-3-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#
```

At the bottom right, there are two buttons: 'Copy' and 'Paste'. At the bottom left, there is a red tab labeled 'Top'.

➤ Configuration PC

Client 1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☒ Static

IPv4 Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::205:5EFF:FEA4:821D

Default Gateway:

DNS Server:

802.1X

☒ Use 802.1X Security

Authentication: MD5

Username:

Password:

Top

1) Configuration du nom, la protection globale et le domaine local

■ Configuration de Telnet

Router1

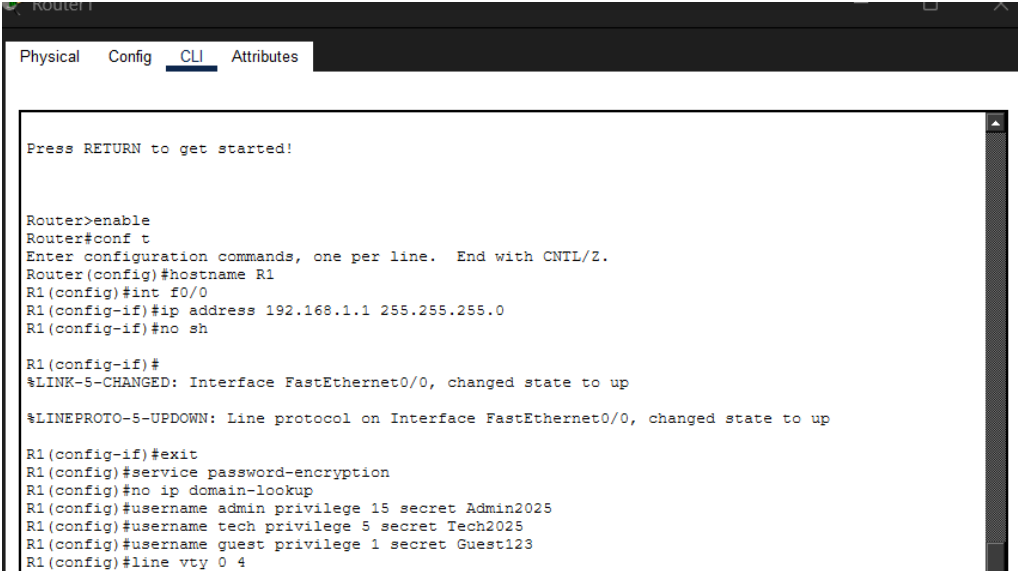
Physical Config **CLI** Attributes

Press RETURN to get started!

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#service password-encryption
```

2) Création des utilisateurs (privilege 1, 5,15)



```
Router1
Physical Config CLI Attributes

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
```

3) Activation de Telnet sur les lignes VTY :



```
Router1
Physical Config CLI Attributes

Press RETURN to get started!

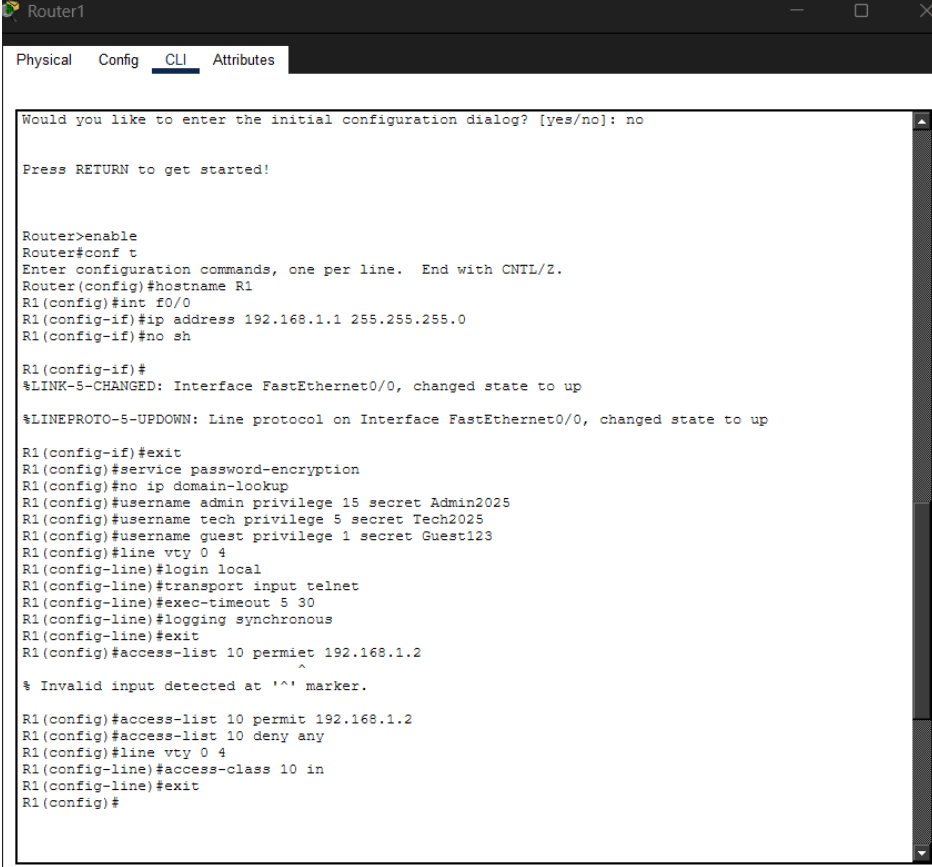
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input telnet
R1(config-line)#exec-timeout 5 30
R1(config-line)#logging synchronous
R1(config-line)#exit
```

4) Configuration du mot de passe de ligne :



```
Router1
Physical Config CLI Attributes

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

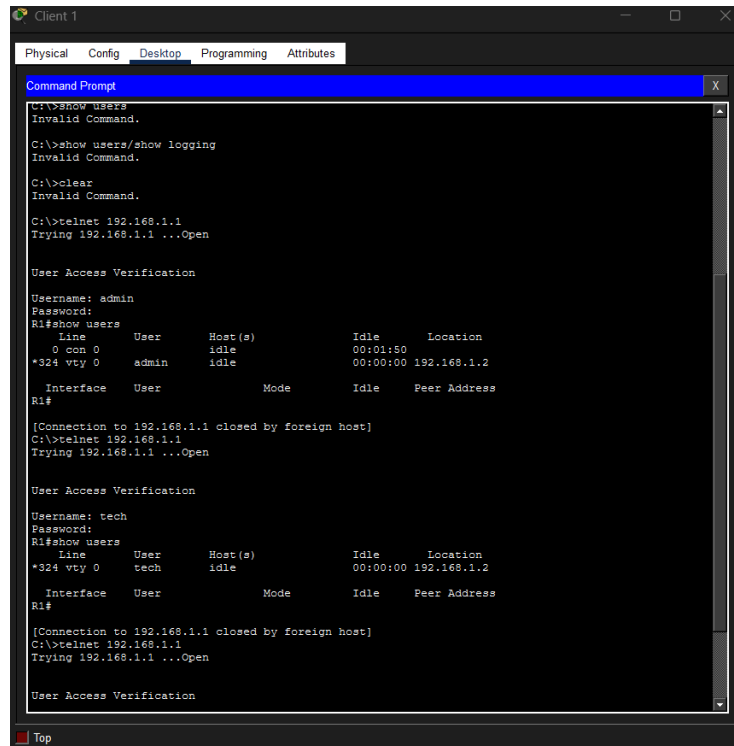
R1(config-if)#exit
R1(config)#service password-encryption
R1(config)#no ip domain-lookup
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest123
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input telnet
R1(config-line)#exec-timeout 5 30
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#access-list 10 permit 192.168.1.2
      ^
% Invalid input detected at '^' marker.

R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#line vty 0 4
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#
```

Copy Paste

Top

5) Test d'accès Telnet :



The screenshot shows a 'Client 1' window in Cisco Packet Tracer. It contains a 'Command Prompt' window with the following text:

```
C:\>show users
Invalid Command.

C:\>show users/show logging
Invalid Command.

C:\>clear
Invalid Command.

C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Username: admin
Password:
R1#show users

   Line    User      Host(s)      Idle    Location
  *324 vty 0   admin      idle        00:01:50  192.168.1.2

   Interface  User      Mode      Idle    Peer Address
R1#

[Connection to 192.168.1.1 closed by foreign host]
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Username: tech
Password:
R1#show users

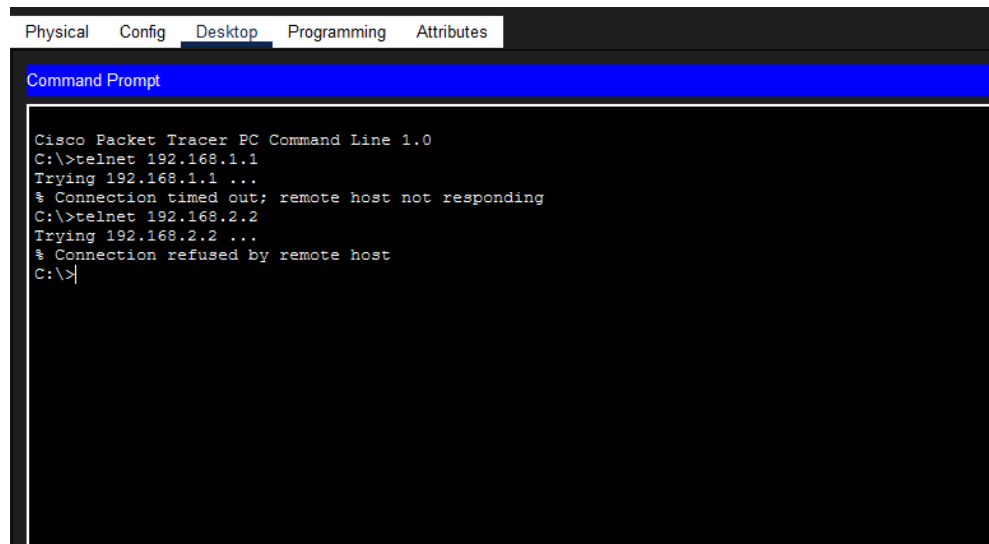
   Line    User      Host(s)      Idle    Location
  *324 vty 0   tech       idle        00:00:00  192.168.1.2

   Interface  User      Mode      Idle    Peer Address
R1#

[Connection to 192.168.1.1 closed by foreign host]
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification
```

PC2 : refuse



The screenshot shows a 'PC Command Line' window in Cisco Packet Tracer. It contains the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...
% Connection timed out; remote host not responding
C:\>telnet 192.168.2.2
Trying 192.168.2.2 ...
% Connection refused by remote host
C:\>
```

Questions

1. Pourquoi Telnet est-il considéré comme non sécurisé ?

R) Telnet est considéré comme non sécurisé parce qu'il envoie toutes les données en clair sur le réseau, sans chiffrement donc toute personne interceptant le trafic peut lire les identifiants, les commandes et les réponses.

2. Quelles informations transitent en clair ?

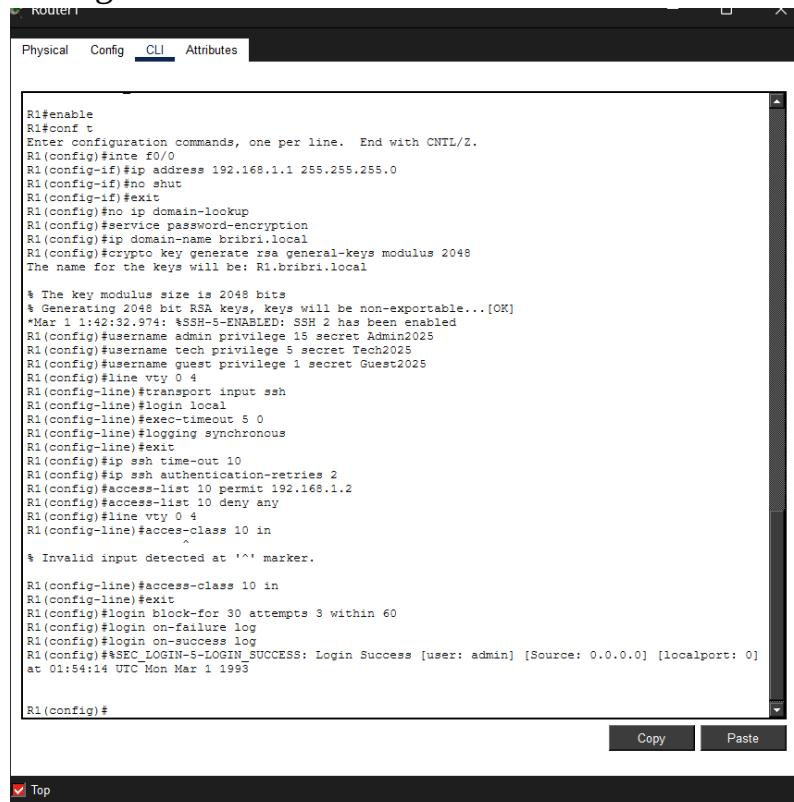
R) Les informations qui transitent en clair sont : le nom utilisateur, le mot de passe, toutes les commandes tapées et les sorties affichées par le routeur.

3. Pourquoi est-il déconseillé d'utiliser Telnet en production ?

R) Il est déconseillé d'utiliser Telnet parce que Telnet expose directement les mots de passe et la configuration aux attaques de type sniffing, ce qui met en danger tout le réseau.

B. Configuration SSH (partie sécurisée)

- Configuration SSH

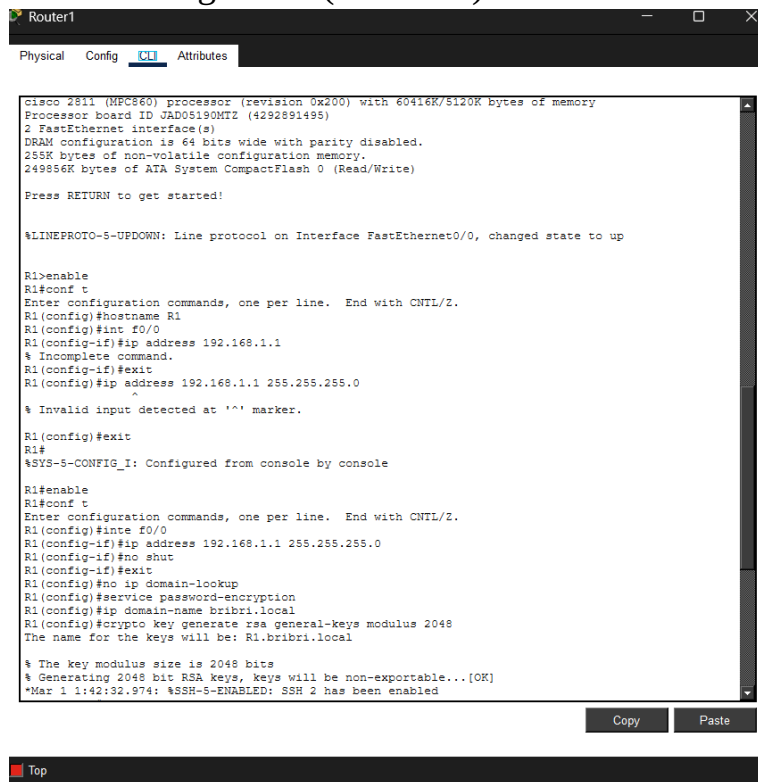


```
R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inte f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name bri bri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.bri bri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest2025
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exec-timeout 5 0
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#line vty 0 4
R1(config-line)#access-class 10 in
R1(config-line)#^
% Invalid input detected at '^' marker.
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#login block-for 30 attempts 3 within 60
R1(config)#login on-failure log
R1(config)#login on-success log
R1(config)#%SEC_LOGIN-5-LOGIN SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0]
at 01:54:14 UTC Mon Mar 1 1993

R1(config)#
```


- Clé de RSA générée (2848 bits)



```
Router1
Physical Config CLI Attributes

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

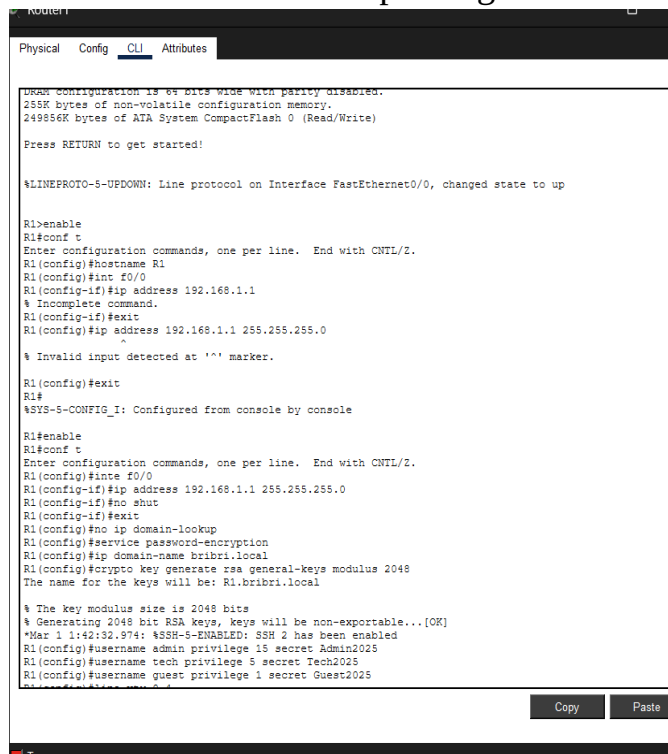
R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1
% Incomplete command.
R1(config-if)#exit
R1(config)#ip address 192.168.1.1 255.255.255.0
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inte f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name bribri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.bribri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
```

- Création des utilisateurs privilégiés



```
Router1
Physical Config CLI Attributes

DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1>enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1
% Incomplete command.
R1(config-if)#exit
R1(config)#ip address 192.168.1.1 255.255.255.0
^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#inte f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#ip domain-name bribri.local
R1(config)#crypto key generate rsa general-keys modulus 2048
The name for the keys will be: R1.bribri.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:42:32.974: %SSH-5-ENABLED: SSH 2 has been enabled
R1(config)#username admin privilege 15 secret Admin2025
R1(config)#username tech privilege 5 secret Tech2025
R1(config)#username guest privilege 1 secret Guest2025
```

- Activation de ssh version 2

```
R1(config)#ip ssh version 2
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
R1(config)#
```

Copy

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Top

- Désactivation Telnet et gardons uniquement SSH :

```
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exec-timeout 5 0
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#ip ssh time-out 10
R1(config)#ip ssh authentication-retries 2
R1(config)#access-list 10 permit 192.168.1.2
R1(config)#access-list 10 deny any
```

Copy

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Top

- Test de l'accès ssh :

✓ Depuis PC1

```

Client 1
Physical Config Desktop Programming Attributes
Command Prompt
Interface      User          Mode          Idle          Peer Address
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 10 secs; Authentication retries: 2
R1#show crypto key mypubkey rsa
% Key pair was generated at: 1:42:32 UTC March 1 1993
Key name: R1.bribri.local
Storage Device: not specified
Usage: General Purpose Key
Key is not exportable.
Key Data:
00003c57 00007b56 00007e67 0000557a 00000e1a 00004ea1 00002205 00002c5b
00007585 00004076 00000f85 00006921 000044a6 000002fb 00007782 000064c7
0000701c 000079a0 00002bbe 000075db 000050cf 00004400 00000e71 7281
% Key pair was generated at: 1:42:32 UTC March 1 1993
Key name: R1.bribri.local.server
Temporary key
Usage: Encryption Key
Key is not exportable.
Key Data:
00002dde 0000700c 00000be0 000013ba 000022d9 00001766 00005232 00001de6
00003631 00006c0b 00003b7f 00007288 000033ec 00007835 00001e2b 00006f4a
0000209d 00001580 00007772 00003e41 00005894 00005f4b 00002443 7b71
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 3 messages logged, xml disabled,
filtering disabled
Monitor logging: disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
filtering disabled

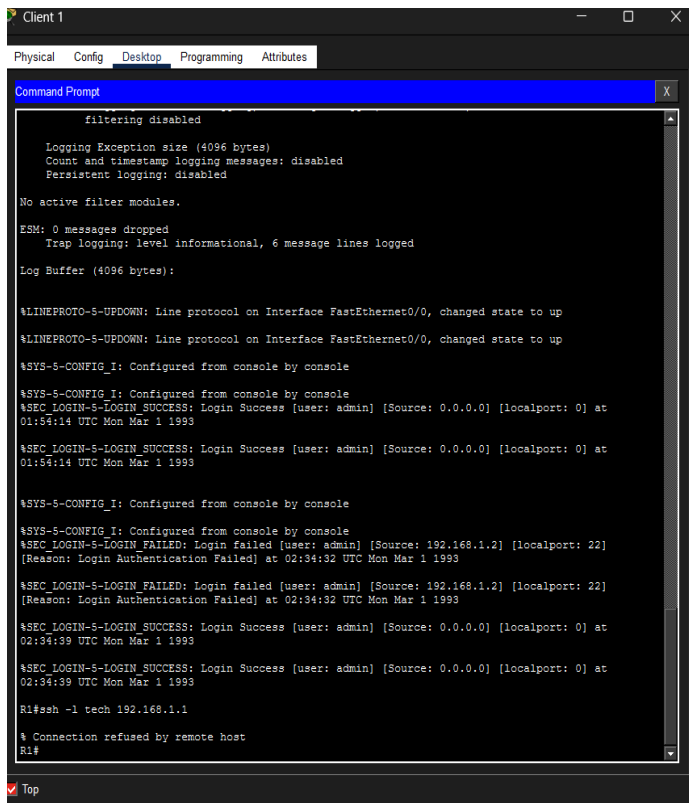
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 3 message lines logged
  
```

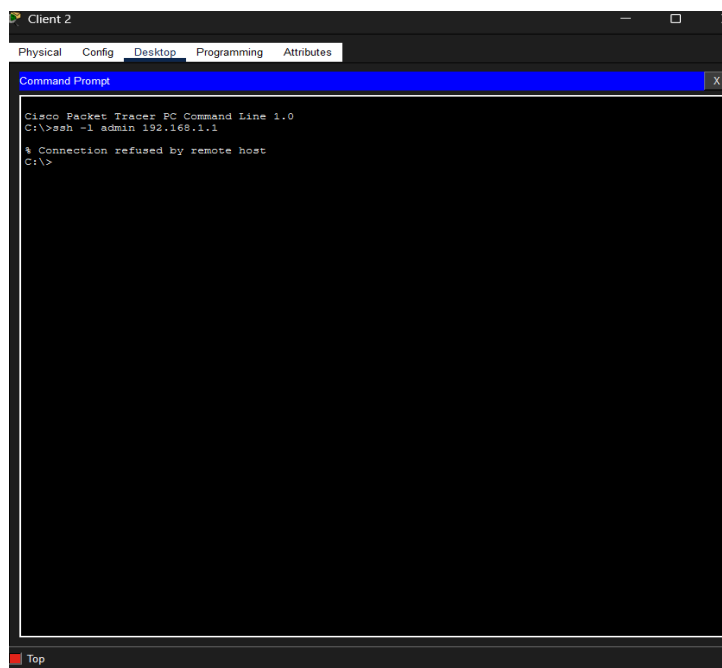
Top

✓ Depuis PC1 en technicien :



```
Client 1
Physical Config Desktop Programming Attributes
Command Prompt
filtering disabled
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled
No active filter modules.
ESM: 0 messages dropped
Trap logging: level informational, 6 message lines logged
Log Buffer (4096 bytes):
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%SYS-5-CONFIG_I: Configured from console by console
%SYS-5-CONFIG_I: Configured from console by console
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
01:54:14 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
01:54:14 UTC Mon Mar 1 1993
%SYS-5-CONFIG_I: Configured from console by console
%SYS-5-CONFIG_I: Configured from console by console
%SEC_LOGIN-5-LOGIN_FAILED: Login failed [user: admin] [Source: 192.168.1.2] [localport: 22]
[Reason: Login Authentication Failed] at 02:34:32 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_FAILED: Login failed [user: admin] [Source: 192.168.1.2] [localport: 22]
[Reason: Login Authentication Failed] at 02:34:32 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
02:34:39 UTC Mon Mar 1 1993
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 0.0.0.0] [localport: 0] at
02:34:39 UTC Mon Mar 1 1993
R1#ssh -l tech 192.168.1.1
% Connection refused by remote host
R1#
```

○ Depuis PC2 : refusé



```
Client 2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.168.1.1
% Connection refused by remote host
C:\>
```

Questions

1. Quelle différence entre SSH v1 et SSH v2 ?

R) SSH v1 est une ancienne version avec des faiblesses de sécurité connues et des algorithmes dépassés tandis que SSH v2 améliore la sécurité, les algorithmes de chiffrement et la gestion des clés, et c'est la seule version recommandée aujourd'hui.

2. Pourquoi RSA 1024 bits n'est plus recommandé ?

R) Parce que aujourd'hui une clé RSA 1024 bits est considérée trop courte par rapport à la puissance de calcul moderne, donc potentiellement cassable. De ce fait on recommande au minimum 2048 bits pour rendre une attaque par force brute ou factorisation irréaliste.

3. Que se passe-t-il si on désactive le domaine local ?

R) Si le domaine est désactivé ou modifié, le routeur peut ne plus avoir de clés valides, ce qui peut empêcher ou perturber les connexions SSH jusqu'à régénération correcte des clés. En effet le domaine local est nécessaire à la génération des clés RSA et au fonctionnement complet de SSH.

Conclusion

Ce TD m'a permis de configurer Telnet et surtout SSH de façon sécurisée sur mon routeur avec les users admin (privilège 15), tech (niveau 5) et guest (1), les ACL pour bloquer tout sauf PC1, les logs des connexions, les timeouts et l'anti-brute-force qui marche nickel. J'ai bien capté pourquoi SSH v2 avec RSA 2048 bits c'est le top (chiffrement total vs Telnet en clair super risqué), et j'ai pu tester les commandes et durcir l'équipement. J'ai pu également tester l'accès de SSH et de Telnet sur les PC.